

Differentiation & Integration

ANSWER KEY



Basic differentiation rules

- 1 Find $\frac{dy}{dx}$ for $y = 3x^4 - 5x^2 + 7x - 2$.
 $\frac{dy}{dx} = 12x^3 - 10x + 7$.
- 2 Find $f'(x)$ for $f(x) = \frac{6}{x^2}$, rewriting first as $6x^{-2}$.
 $f(x) = 6x^{-2}$, so $f'(x) = -12x^{-3} = -\frac{12}{x^3}$.
- 3 Find $\frac{dy}{dx}$ for $y = 5x^3 + \frac{2}{x} - 4$, rewriting the second term first.
Rewrite $y = 5x^3 + 2x^{-1} - 4$. $\frac{dy}{dx} = 15x^2 - 2x^{-2} = 15x^2 - \frac{2}{x^2}$.

Gradient of a curve at a point

- 4 Find the gradient of $y = x^3 - 4x$ at the point where $x = 2$.
 $\frac{dy}{dx} = 3x^2 - 4$. At $x = 2$: $3(4) - 4 = 8$.
- 5 Find the gradient of $y = 2x^3 + x^2 - 5$ at the point where $x = -1$.
 $\frac{dy}{dx} = 6x^2 + 2x$. At $x = -1$: $6(1) + 2(-1) = 6 - 2 = 4$.

Tangents and normals

- 6 Find the equation of the tangent to $y = x^2 - 3x + 5$ at the point $(2, 3)$.
 $\frac{dy}{dx} = 2x - 3$. At $x = 2$: gradient = 1. Tangent: $y - 3 = 1(x - 2)$, so $y = x + 1$.
- 7 Find the equation of the normal to $y = x^2$ at the point $(1, 1)$.
 $\frac{dy}{dx} = 2x$. At $x = 1$: tangent gradient = 2, so normal gradient = $-\frac{1}{2}$. Normal: $y - 1 = -\frac{1}{2}(x - 1)$, so $y = -\frac{1}{2}x + \frac{3}{2}$.
- 8 Find the equation of the tangent to $y = x^3 - 2x + 1$ at the point $(1, 0)$.
 $\frac{dy}{dx} = 3x^2 - 2$. At $x = 1$: gradient = 1. Tangent: $y - 0 = 1(x - 1)$, so $y = x - 1$.

Stationary points

- 9 Find the coordinates of the stationary point of $y = x^2 - 6x + 11$, and determine whether it is a maximum or minimum.
 $\frac{dy}{dx} = 2x - 6 = 0$, so $x = 3$. $y = 9 - 18 + 11 = 2$. Stationary point: $(3, 2)$. Since $\frac{d^2y}{dx^2} = 2 > 0$, it is a minimum.

- 10 Find the stationary points of $y = x^3 - 3x^2 - 9x + 5$, and classify each using the second derivative.
 $\frac{dy}{dx} = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x - 3)(x + 1) = 0$, so $x = 3$ or $x = -1$. $\frac{d^2y}{dx^2} = 6x - 6$. At $x = 3$: $12 > 0$, a minimum. At $x = -1$: $-12 < 0$, a maximum.
- 11 Find the stationary points of $y = x^3 - 12x + 2$, and classify each using the second derivative.
 $\frac{dy}{dx} = 3x^2 - 12 = 3(x - 2)(x + 2) = 0$, so $x = 2$ or $x = -2$. $y(2) = 8 - 24 + 2 = -14$;
 $y(-2) = -8 + 24 + 2 = 18$. $\frac{d^2y}{dx^2} = 6x$. At $x = 2$: $12 > 0$, a minimum at $(2, -14)$. At $x = -2$:
 $-12 < 0$, a maximum at $(-2, 18)$.

Basic integration

- 12 Find $\int (4x^3 - 6x + 3) dx$.
 $\int (4x^3 - 6x + 3) dx = x^4 - 3x^2 + 3x + C$.
- 13 Find $\int \frac{1}{x^3} dx$, rewriting first as x^{-3} .
 $\int x^{-3} dx = \frac{x^{-2}}{-2} + C = -\frac{1}{2x^2} + C$.
- 14 Find $\int (3x^2 + 4x - 5) dx$.
 $\int (3x^2 + 4x - 5) dx = x^3 + 2x^2 - 5x + C$.

Finding a curve given its gradient

- 15 A curve has $\frac{dy}{dx} = 6x^2 - 4x$ and passes through $(1, 3)$. Find the equation of the curve.
 $y = 2x^3 - 2x^2 + C$. Using $(1, 3)$: $3 = 2 - 2 + C$, so $C = 3$. Equation: $y = 2x^3 - 2x^2 + 3$.
- 16 A curve has $\frac{dy}{dx} = 3x^2 + 2$ and passes through $(2, 15)$. Find the equation of the curve.
 $y = x^3 + 2x + C$. Using $(2, 15)$: $15 = 8 + 4 + C$, so $C = 3$. Equation: $y = x^3 + 2x + 3$.

Definite integrals and the area under a curve

- 17 Evaluate $\int_1^3 (2x + 1) dx$.
 $\int_1^3 (2x + 1) dx = [x^2 + x]_1^3 = (9 + 3) - (1 + 1) = 12 - 2 = 10$.
- 18 Evaluate $\int_0^2 (3x^2 - 2x) dx$.
 $\int_0^2 (3x^2 - 2x) dx = [x^3 - x^2]_0^2 = (8 - 4) - (0 - 0) = 4$.

Using integration to find the area between a curve and the x-axis

- 19** Find the area enclosed between the curve $y = 4 - x^2$ and the x -axis, between its two zeros.

Zeros: $4 - x^2 = 0$, so $x = \pm 2$. Area

$$= \int_{-2}^2 (4 - x^2) dx = \left[4x - \frac{x^3}{3} \right]_{-2}^2 = \left(8 - \frac{8}{3} \right) - \left(-8 + \frac{8}{3} \right) = 16 - \frac{16}{3} = \frac{32}{3}.$$

- 20** Find the area enclosed between the curve $y = 9 - x^2$ and the x -axis, between its two zeros.

Zeros: $9 - x^2 = 0$, so $x = \pm 3$. Area

$$= \int_{-3}^3 (9 - x^2) dx = \left[9x - \frac{x^3}{3} \right]_{-3}^3 = (27 - 9) - (-27 + 9) = 18 - (-18) = 36.$$

A well-designed word problem: minimizing material for a can

- 21** This question has two parts. A cylindrical can must have a volume of $500\pi \text{ cm}^3$. Let r be the radius and h the height. (a) Show that the surface area is $S = 2\pi r^2 + \frac{1000\pi}{r}$. (b) Find the radius that minimizes the surface area, and confirm it is a minimum using the second derivative.

(a) Volume: $\pi r^2 h = 500\pi$, so $h = \frac{500}{r^2}$. Surface area (two circular ends plus the curved side):
 $S = 2\pi r^2 + 2\pi r h = 2\pi r^2 + 2\pi r \left(\frac{500}{r^2} \right) = 2\pi r^2 + \frac{1000\pi}{r}$. (b) $\frac{dS}{dr} = 4\pi r - \frac{1000\pi}{r^2} = 0$, so $4\pi r = \frac{1000\pi}{r^2}$, giving $r^3 = 250$, so $r = \sqrt[3]{250} \approx 6.30 \text{ cm}$. Second derivative: $\frac{d^2S}{dr^2} = 4\pi + \frac{2000\pi}{r^3}$, which is always positive for $r > 0$, confirming this is a minimum.